#2 Graph K_5 has 5 vertices and exactly 10 edges (excluding loops): here we assume that `graph.simpleEdges` returns all edges in a graph excluding loops.

Listing 15.4 shows the method to check if a graph is (isomorphic to) K_5: obviously it needs to have 5 vertices (!), but we also know how many edges it should have, exactly 10; notice that here we are talking about simple edges, where source and destination are distinct (so, no loops).

Listing 15.5 Planarity Testing utility methods

```
function
isK3_3(graph)                                                    #1
  (n,m) ← (|graph.vertices|,
|graph.simpleEdges|)                      #2
  if n == 6 and m == 9  then
         #3
    if isBipartite(graph) and partitionsSize(graph) == (3,3)
then    #4
      return true
  return false
```

#1 Method `isK3-3` takes a graph and checks if it is the complete bipartite graph with 2 partitions of three vertices each.
#2 Temporary variables for number of vertices and edges in `graph`.
#3 Graph $K_{3,3}$ has 6 vertices and exactly 9 edges (excluding loops): here we assume that `graph.simpleEdges` returns all edges in a graph excluding loops.